

BUILD WITH AI: CODE FOR COMMUNITIES — SECOND EDITION

ArogyaNexus AI

Prototype evaluation for Smart Health & Supply Chain Resilience

GTC Labs

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Problem Statement 03 · BRICS Resilience

A PHC intelligence command center

From prototype to jury decision

The opportunity

- ♦ Why PHC supply chains fail under pressure
- ♦ How the hackathon challenge defines success
- ♦ Where ArogyaNexus fits

The evaluation

- ♦ Live prototype evidence
- ♦ Google AI integration
- ♦ Impact, scalability, risks, and score

Jury question: Does this prototype turn frontline health signals into faster, safer resource decisions?

When a PHC runs out, the whole response slows down

Public healthcare systems across developing nations face persistent supply-chain vulnerabilities. Limited real-time visibility into medicines, patient footfall, and resources leads to stock-outs when response capacity matters most.

Visibility gap

Medicine stocks, beds, staff, and patient demand are fragmented across PHCs.

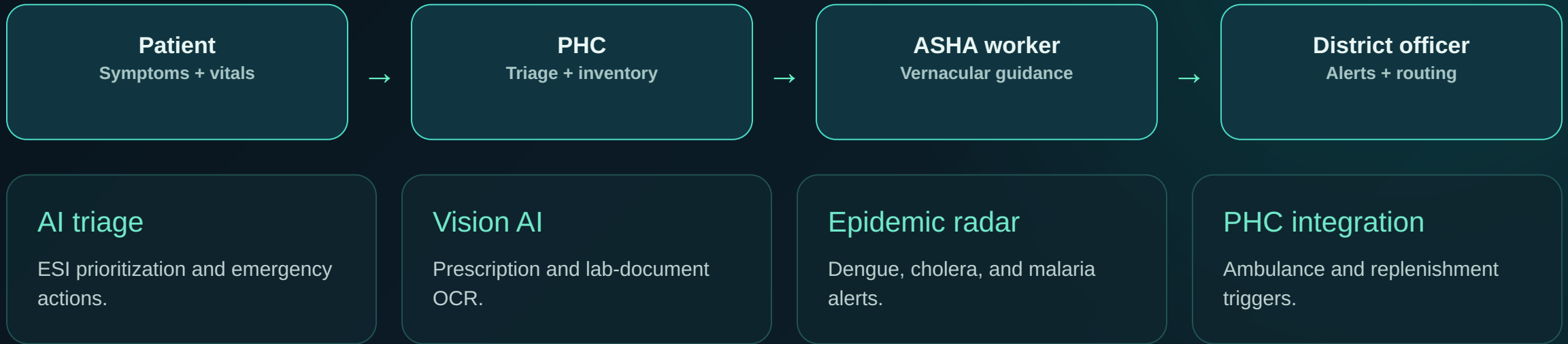
Prediction gap

Emergency surges are detected after resources are already under pressure.

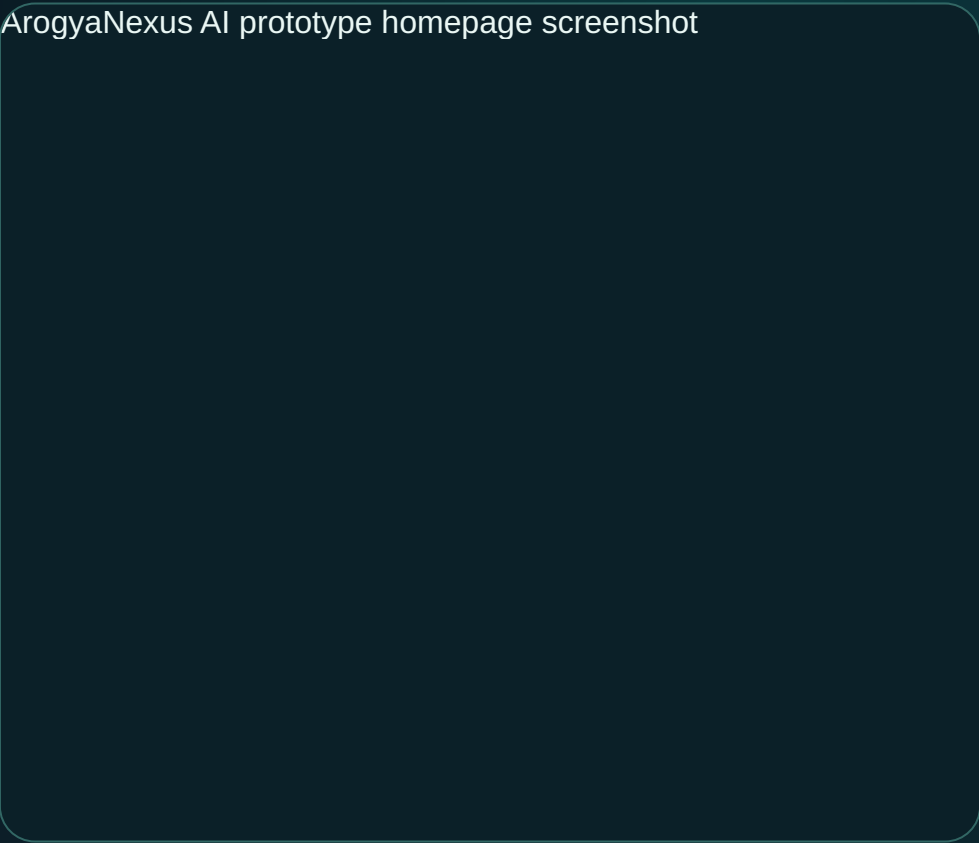
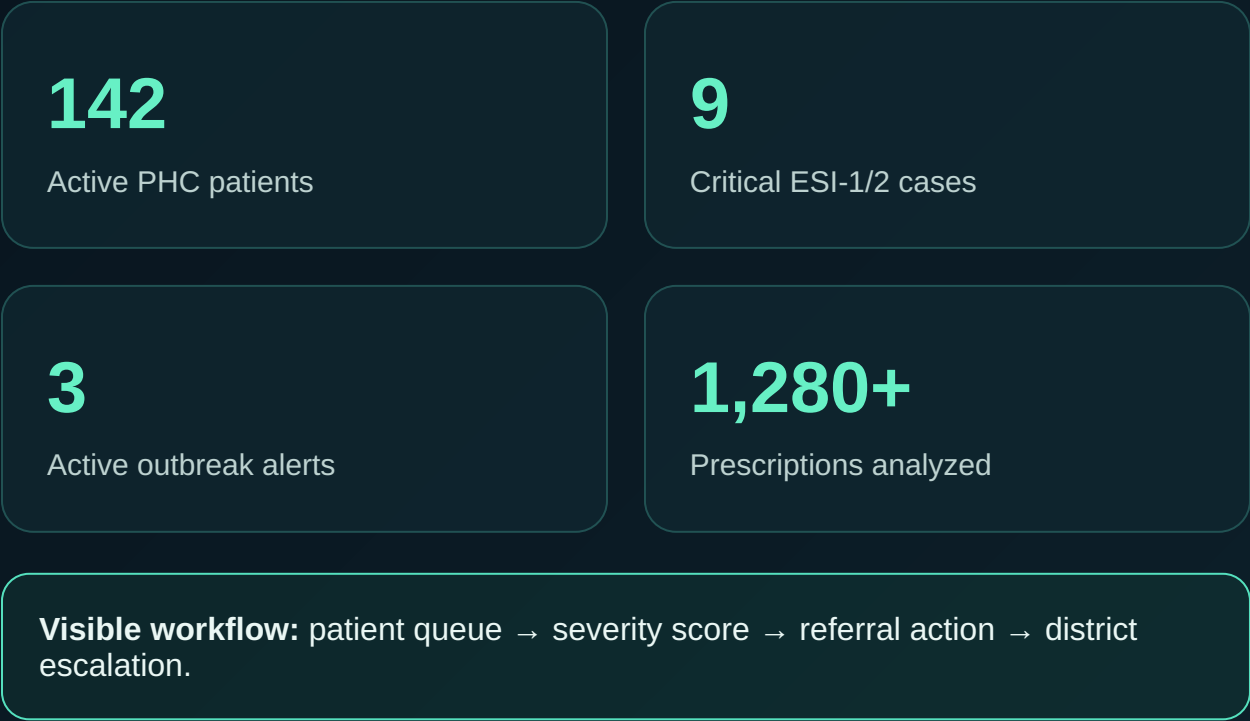
Coordination gap

Districts lack a shared, auditable way to redistribute scarce resources.

ArogyaNexus connects the PHC response loop



A working dashboard, not only a concept



Gemini multimodal intelligence at the point of care

Evidence on the public prototype

- ♦ Google Gemini 1.5 Multimodal Vision is explicitly named.
- ♦ Gemini 1.5 Flash Vision is named for prescription analysis.
- ♦ The scanner extracts medicine names, dosages, timing, and duration.
- ♦ Vernacular explanations are presented in Hindi and Telugu.

Jury interpretation

11 / 20

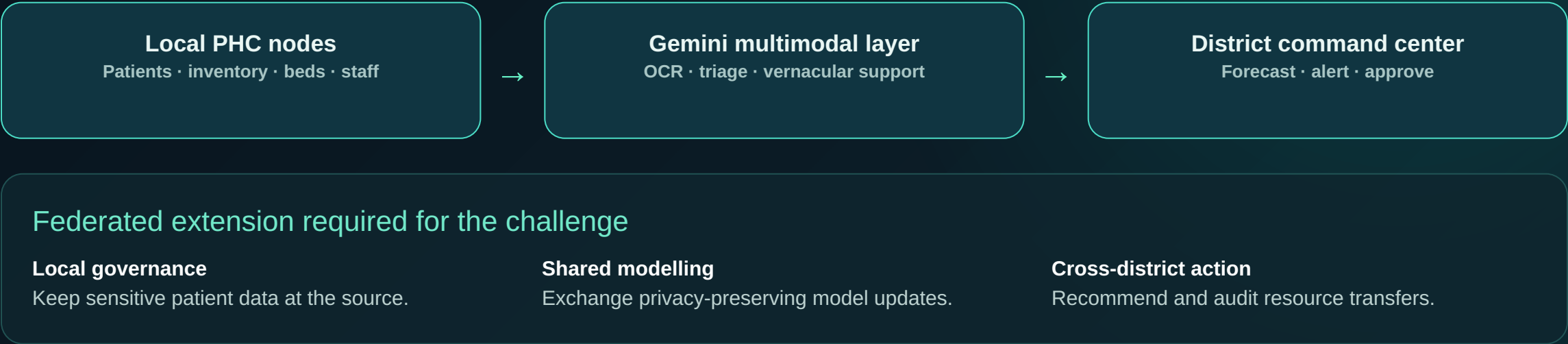
Google AI integration depth



Strong visible Gemini layer. AI Studio, Vertex AI, BigQuery, Firebase, Maps Platform, and Cloud Run are not evidenced on the public prototype.

Claim discipline: Present non-evidenced Google services as planned integrations, not as completed implementation. Confirm Google AI Studio and Vertex AI credit details with organizers.

From local PHC signals to federated resilience



Promising prototype with one major gap

Strengths

- ♦ High-relevance public-health problem
- ♦ Concrete multi-workflow prototype
- ♦ Visible Gemini vision integration
- ♦ Clear PHC and ASHA use cases
- ♦ Strong emergency-response narrative

Indicative assessment

16/20

Innovation

13/20

Technical

15/20

Impact & scale

12/15

UX



75 / 100

Indicative reviewer score, not official organizer scoring.

The value is measured in response time and avoided stock-outs

For patients

- ♦ Faster severity recognition
- ♦ Earlier referral and ambulance routing
- ♦ Clearer medicine instructions

For PHCs

- ♦ Demand and outbreak signals
- ♦ O2 and anti-venom replenishment triggers
- ♦ ASHA-friendly field workflows

For districts

- ♦ Syndromic cluster alerts
- ♦ Resource redistribution decisions
- ♦ Auditable emergency response

Scale pathway: pilot PHCs → multi-district network → national PHC command center → BRICS shared predictive modelling.

What the jury will challenge

Data consistency

Aarav Sharma appears as age 6 on the homepage and age 26 on the triage page. Fix before judging.

Clinical safety

Treatment suggestions need clinician verification, contraindication checks, confidence, and audit logs.

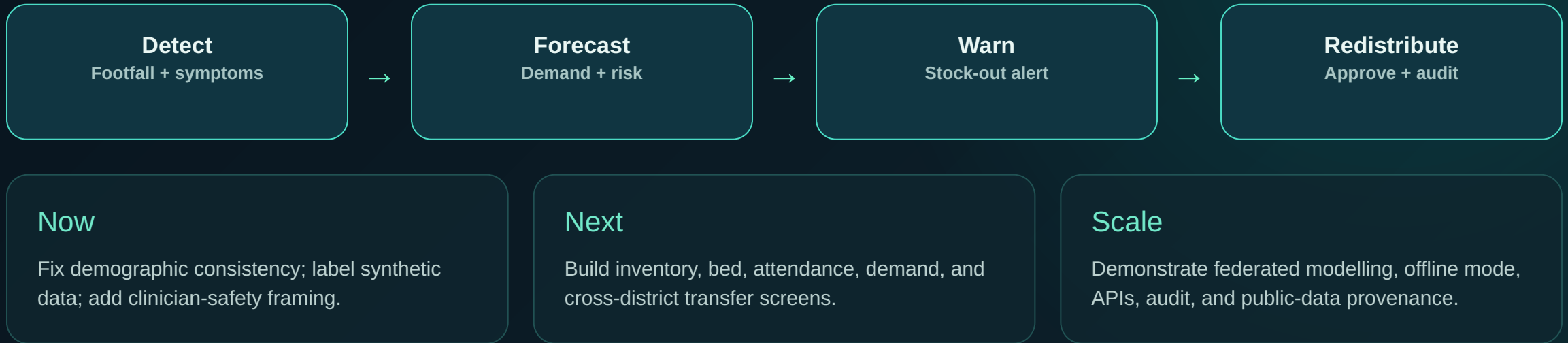
Evidence quality

The 97.4% scanner figure needs a dataset, methodology, error analysis, and language breakdown.

Supply-chain proof

Add visible screens for stock, beds, staff, demand forecasts, stock-out warnings, and redistribution.

The next demo should close the challenge loop



Why GTC Labs can win

- ♦ A credible clinical foundation for a high-impact public-health problem.
- ♦ A visible Gemini multimodal AI layer, not just a concept.
- ♦ A clear path from frontline signals to district resource decisions.
- ♦ A strong resilience story: forecast, warn, redistribute, audit.

The ask

Help us turn every PHC signal into an earlier, safer, and more coordinated health-resource decision.

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Prototype: <https://arogyanexus-ai.onrender.com/>
Hackathon: <https://hack2skill.com/event/codeforcommunities2>